

ATOMS

Introduction

Definition:

- Atoms are the basic building blocks of matter.
- An atom consists of electrons, protons and neutrons.

Thomson's Atomic Model

Statement:

According to Thomson, an atom is a positively charged sphere with electrons embedded in it.

Limitation:

This model could not explain the scattering experiment results.

Rutherford's Atomic Model

Statement:

According to Rutherford, most of the mass and positive charge of an atom is concentrated at the center called the nucleus.

Observations:

- Most alpha particles passed straight.
- Few particles were deflected.
- Very few bounced back.

Bohr's Atomic Model

Postulates:

- Electrons revolve around the nucleus in fixed circular orbits.
- Electrons do not radiate energy while revolving.
- Each orbit has fixed energy.
- Energy is emitted or absorbed when an electron jumps between orbits.

Radius of Orbit

Formula:

$$r_n = 0.53 n^2 \text{ \AA}$$

Where:

- r_n = Radius of nth orbit
- n = Principal quantum number

Energy of Electron

Formula:

$$E_n = -13.6 / n^2 \text{ eV}$$

Where:

- E_n = Energy of electron
- n = Orbit number

Important Point:

Energy of electrons increases with increase in orbit number.

Hydrogen Spectrum

Definition:

The spectrum obtained due to the transition of electrons in the hydrogen atom is called the hydrogen spectrum.

Spectral Series

- Lyman Series
- Balmer Series
- Paschen Series
- Brackett Series
- Pfund Series

Bohr's Explanation

When an electron jumps from higher energy orbit to lower energy orbit, energy is emitted in the form of radiation.

Energy Relation:

$$\Delta E = h\nu$$

Where:

- ΔE = Energy difference
- h = Planck's constant
- ν = Frequency of radiation

Rydberg Formula

$$1/\lambda = R[(1/n_1^2) - (1/n_2^2)]$$

Where:

- λ = Wavelength
- R = Rydberg constant
- n_1, n_2 = Orbit numbers

Applications of Atomic Models

- Study of atomic structure
- Explanation of hydrogen spectrum
- Development of quantum theory
- Understanding electron transitions

Limitations of Bohr's Model

- Applicable only for hydrogen atoms.
- Could not explain the fine spectrum.
- Could not explain the Zeeman effect.
- Failed to explain multi-electron atoms.

Quick Revision Points

- Atoms are basic units of matter.
- Rutherford proposed a nuclear model.
- Bohr proposed fixed energy orbits.

- Energy is emitted during electron transition.
- Hydrogen spectrum consists of spectral series.
- $\Delta E = h\nu$

Important Formulae

$$r_n = 0.53 n^2 \text{ \AA}$$

$$E_n = -13.6 / n^2 \text{ eV}$$

$$\Delta E = h\nu$$

$$1/\lambda = R[(1/n_1^2) - (1/n_2^2)]$$